Sanjay M

Vellore, Tamil Nadu, India

Sanjay M-LinkedIn | https://github.com/Sanjay210304 | sanjay217010@gmail.com | +91 7010726700

Education

Vellore Institute of Technology,Vellore. (Integrated Mtech, Software Engineering) Sep/2021–July/2026

Class XII(Sunbeam School, Vellore) June/2020–May/2021

Class X (Koona School, Vellore) June/2018–May/2019

Academic Projects

Time Series Analysis with Cryptocurrency Data July/2024 - Dec/2024

- Developed an LSTM-based deep learning model for cryptocurrency price prediction, optimizing accuracy through advanced feature engineering. Evaluated model performance using key metrics to ensure reliable financial forecasting.
- Designed and implemented automated pipelines for data preprocessing, visualization, and trend analysis. Utilized Python, TensorFlow, Pandas, and Plotly to streamline financial data insights.

MiniLang – LLVM-Based Toy Language with JIT Compilation Sep/2024 - Feb/2025

- Designed and implemented a statically-typed toy language with custom syntax, including parsing, semantic analysis, and code generation phases.
- Compiled source code to LLVM IR using llvmlite, enabling Just-In-Time execution and demonstrating end-to-end understanding of modern compiler backends.

Serverless Student Data Management System on AWS June/2023 - Nov/2023

- Developed and deployed a serverless student data management system using AWS services (S3, API Gateway, Lambda, DynamoDB, and CloudFront) for efficient and scalable data handling.
- Implemented RESTful-APIs with AWS Lambda (Python) to perform secure CRUD operations on DynamoDB, optimizing performance with pagination and API Gateway security measures.

Smart Biogas Production with IoT and Machine Learning June/2024 - Nov/2024

- Developed an IoT-integrated biogas monitoring system utilizing sensors and machine learning techniques to optimize gas yield and enable predictive maintenance. Implemented real-time data analytics and anomaly detection with MATLAB and ThingSpeak.
- Designed and deployed a smart biogas production system that leverages IoT technology and machine learning algorithms to monitor gas output and predict maintenance needs. Utilized MATLAB and ThingSpeak for real-time data analysis and anomaly detection.

Custom MLIR Dialect for Math Rule Optimization Jan/2025 - April/2025

- Developed a custom MLIR dialect to represent mathematical and fraud-detection rules with tailored operations, types, and attributes for domain-specific optimization.
- Implemented transformation and lowering passes to Affine and LLVM dialects, showcasing deep understanding of MLIR's modular IR design and compiler backend infrastructure.

Fraud Analyst Copilot: A Retrieval-Augmented and Explainable AI System for Financial Fraud Detection" Jan/2025 - April/2025

- Built a Fraud Analyst Copilot using Retrieval-Augmented Generation (RAG) with SentenceTransformers, FAISS, and OpenAI GPT-4 to generate context-aware fraud insights from historical credit card transactions.
- Implemented explainable fraud detection using CatBoost and SHAP, enabling interpretable model predictions and real-time visualizations of feature contributions for enhanced analyst decision support.

Skills

- **Programming Languages**: Python, C, C++.
- **Deep Learning Frameworks**: Tensorflow, Pytorch, JAX.
- **Compiler Tools**: LLVM, MLIR, GCC, Flex/Bison, LLvmlite, Clang
- **Build Systems**: CMake, Ninja
- **Machine Learning Libraries**: Pandas, Ploty, Matplot, Numpy, Scikit-Learn, Keras.
- **Relevant Coursework**:Operating System, Computer Networks, Data Structures and Algorithm, OOPS, DBMS
- **PEFT Methods**: RLHF, DPO, LORA, DORA.

Certifications

- **C,C++** : Udemy.